

Supplement S2

Replication code for the article “Conspiracy theories are central to the complex system of predictors of Italian vaccine hesitancy”

This document provides the replication code for all the analyses shown in the article and in Supplement S1. This is Rmarkdown code knitted in a Word file. All the documentation is already available in a GitHub folder. Code follows the IPO [Input, Processing, Output] protocol and is fully reproducible. However, code is shared in this way to ensure anonymity. After the peer review, the GitHub link could be inserted directly into the main article.

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File 1: Data Management

Libraries

```
#packages
library("pacman")
p_load(tidyverse, here, sjlabelled, stringr, glue, janitor, haven, stargazer,
       ltm, skimr, readxl)

#remove scientific notation
options(scipen=999)
```

Input

```
#Load database
response_original = read_rds(here("Input", "v1.0_ResPOnsE_COVID_19_W1-W4-2
.rds")) %>%
  clean_names()
```

Processing

W3

```
# 10 November to 22 December 2021
#Select and rename variables
W3 = response_original %>%
  filter(info_wave==3) %>%
  dplyr::select(c(v2,v3,j5_01,v1,f10,f3,f9,h1_04,j4bis_b,b2_03,b2_06,b2_07,d5,d1,
                 e4_bis,k3_03,k3_04,k3_05,e2_01,e2_04,j5_02,g7,c5,s1,s2,
                 s9,s8,c1)) %>%
  mutate(across(v2:c1, ~replace(., .>97 , NA))) %>%
  na.omit()

#colnames
colnames(W3) = c("vac", "vac_int", "vac_bad", "vac_ob", "worry", "risk", "conspiracy",
                 "nat", "h_locus", "comp_dist", "comp_mask", "comp_hand", "heco",
                 "judg_gov_covid", "judg_gov", "PTV_L", "PTV_5SM", "PTV_BOI", "tr_par",
                 "tr_EU", "tr_sci", "pray", "media", "sex", "age", "educ",
                 "reg", "eco_insec")

#combine vac_int and vac in INT_VAC
W3 = W3 %>%
  mutate(hesitancy = case_when(
    (vac_int < 3 | vac == 1)~ 0,
    (vac_int > 2 & vac == 2)~ 1,)) %>%
  dplyr::select(-c(vac_int, vac))

#invert polarity and recode
W3 = W3 %>%
```

```
mutate((across(vac_bad, ~ 6 - .)),
worry = ifelse(worry<=2, 1, 0),
conspiracy = ifelse(conspiracy<3, 1, 0),
pray = ifelse(pray<=4, 0, 1),
media = ifelse(media>=4 & media<=7, 1, 0),
sex = ifelse(sex==2, 1, 0),
educ = ifelse(educ<=5, 1, 0),
(across(age, ~ 89 - .)),
reg = ifelse(reg<=3, 1, 0),
(across(comp_dist:comp_hand, ~ 10 - .)),
(across(judg_gov_covid:judg_gov, ~ 10 - .)),
(across(tr_par:tr_EU, ~ 10 - .)))
```

Inspect true number of missing

```
#Load valid cases spreadsheet (available at: https://dataverse.unimi.it/data
set.xhtml?persistentId=doi:10.13130/RD_UNIMI/FF0ABQ)
valid = read_excel(here("Input", "Valid_cases.xlsx"))
```

```
#Select V I used in the analysis to retrieve true number of N
```

```
valid = valid %>%
  dplyr::select(c(v2,v3,j5_01,v1,f10,f3,f9,h1_04,j4bis_b,b2_03,b2_06,b2_07,d5,d1,
                  e4_bis,k3_03,k3_04,k3_05,e2_01,e2_04,j5_02,g7,c5,s1,s2,
                  s9,s8,c1))
```

```
#Extract minimum number of obs per question
```

```
min(valid)
```

```
## [1] 3767
```

```
# TRUE N = 3767. Thus I had 3767 before listwise, 1535 after. It means I m
working with 40.7% of
# the original sample.
```

PCA

```
#PCA
```

```
#compliance
```

```
compliance = data.frame(W3$comp_mask, W3$comp_hand, W3$comp_dist)
```

```
compliance = na.omit(compliance)
```

```
pr_compliance <- princomp(na.omit(compliance), cor = TRUE)
```

```
W3$low_comp = apply(W3[8:10], 1, mean, na.rm = TRUE)
```

```
W3 = W3 %>%
```

```
  dplyr::select(-c(comp_hand, comp_mask, comp_dist))
```

```
#Gov
```

```
gov = data.frame(W3$judg_gov_covid, W3$judg_gov)
```

```
gov = na.omit(gov)
```

```
pr_gov <- princomp(na.omit(gov), cor = TRUE)
```

```
W3$distrust_gov = apply(W3[9:10], 1, mean, na.rm = TRUE)
```

```
W3 = W3 %>%
```

```
  dplyr::select(-c(judg_gov_covid, judg_gov))
```

```

#inst_tr
inst_tr = data.frame(W3$tr_par, W3$tr_EU)
inst_tr = na.omit(inst_tr)
pr_inst_tr <- princomp(na.omit(inst_tr), cor = TRUE)

W3$distrust_inst = apply(W3[12:13], 1, mean, na.rm = TRUE)
W3 = W3 %>%
  dplyr::select(-c(tr_par, tr_EU))

```

Output

[descriptives \(Article\)](#)

```

summary_vaccine = as.data.frame(skim(W3))
stargazer(as.data.frame(W3), type = "text", nobs = TRUE)

```

```

##
## =====
## Statistic      N      Mean  St. Dev.  Min    Max
## -----
## vac_bad        1,540  2.148   1.244     1     5
## vac_ob         1,540  2.401   1.325     1     5
## worry          1,540  0.673   0.469     0     1
## risk           1,540  3.173   0.756     1     5
## conspiracy      1,540  0.456   0.498     0     1
## nat            1,540  4.142   3.187     0    10
## h_locus        1,540  6.453   2.586     0    10
## he_eco         1,540  4.911   2.642     0    10
## PTV_L          1,540  2.562   3.504     0    10
## PTV_5SM        1,540  2.692   3.274     0    10
## PTV_BOI        1,540  2.525   3.460     0    10
## tr_sci         1,540  2.252   1.163     1     5
## pray          1,540  0.550   0.498     0     1
## media          1,540  0.465   0.499     0     1
## sex            1,540  0.454   0.498     0     1
## age            1,540  41.744  14.895     1    71
## educ           1,540  0.584   0.493     0     1
## reg            1,540  0.433   0.496     0     1
## eco_insec      1,540  2.506   0.640     1     4
## hesitancy      1,540  0.171   0.377     0     1
## low_comp       1,540  1.281   1.806    0.000  10.000
## distrust_gov   1,540  4.403   2.264    0.000  10.000
## distrust_inst  1,540  5.521   2.465    0.000  10.000
## -----

```

[Summary of PCAs \(Supplement S1 section 2\)](#)

#dimensionality

pr_compliance *#unidimensional*

```

## Call:
## princomp(x = na.omit(compliance), cor = TRUE)
##
## Standard deviations:

```

```

##      Comp.1      Comp.2      Comp.3
## 1.5009023 0.6149379 0.6075720
##
## 3 variables and 1540 observations.

pr_gov #unidimensional

## Call:
## princomp(x = na.omit(gov), cor = TRUE)
##
## Standard deviations:
##      Comp.1      Comp.2
## 1.3608960 0.3846584
##
## 2 variables and 1540 observations.

pr_inst_tr #unidimensional

## Call:
## princomp(x = na.omit(inst_tr), cor = TRUE)
##
## Standard deviations:
##      Comp.1      Comp.2
## 1.2820125 0.5970292
##
## 2 variables and 1540 observations.

#alphas
cronbach.alpha(compliance, CI=TRUE, standardized=TRUE)

##
## Standardized Cronbach's alpha for the 'compliance' data-set
##
## Items: 3
## Sample units: 1540
## alpha: 0.834
##
## Bootstrap 95% CI based on 1000 samples
## 2.5% 97.5%
## 0.805 0.859

cronbach.alpha(gov, CI=TRUE, standardized=TRUE)

##
## Standardized Cronbach's alpha for the 'gov' data-set
##
## Items: 2
## Sample units: 1540
## alpha: 0.92
##
## Bootstrap 95% CI based on 1000 samples
## 2.5% 97.5%
## 0.908 0.932

cronbach.alpha(inst_tr, CI=TRUE, standardized=TRUE)

```

```
##  
## Standardized Cronbach's alpha for the 'inst_tr' data-set  
##  
## Items: 2  
## Sample units: 1540  
## alpha: 0.783  
##  
## Bootstrap 95% CI based on 1000 samples  
## 2.5% 97.5%  
## 0.753 0.808
```

save

#export data

```
saveRDS(W3, here("Input", "W3.rds"))
```

File 2: Analyses (Output informs both Article and Supplement S1)

```
library(pacman)
p_load(tidyverse, here, sjlabelled, stringr, glue, EGAnet, janitor, haven,
       ggpubr, gridExtra, dplyr, GGally, qgraph, sjmisc, igraph, sjPlot,
       grid, psych, stargazer, mgm, backbone, ggplot2, tnet, ggrepel,
       NetworkComparisonTest, Matrix, bootnet, matrixcalc, openxlsx, devtools, jtools)
```

Input

```
W3 = readRDS((here("Input", "W3.rds")))
```

Processing

Network objects

network objects

```
type <- c("g", "g", "c", "g", "c", "g", "g", "g", "g", "g", "g", "g", "g", "c", "c", "c", "g", "c",
         "c", "g", "c", "g", "g", "g")
```

```
level <- c("1", "1", "2", "1", "2", "1", "1", "1", "1", "1", "1", "1", "1", "2", "2", "2", "1",
          "2", "2", "1", "2", "1", "1", "1")
```

```
shortnames <- c(
  "vac_bad",
  "vac_free",
  "low_worry",
  "low_risk",
  "conspiracy",
  "nat",
  "int_locus",
  "low_col_resp",
  "PTV_L",
  "PTV_5SM",
  "PTV_BOI",
  "distrust_sci",
  "pray",
  "media",
  "female",
  "young",
  "low_educ",
  "rural",
  "eco_insec",
  "hesitancy",
  "low_comp",
  "distrust_gov",
  "distrust_inst")
```

```
longnames <- c(
  "Vaccine bad for health",
  "Against mandatory vaccination",
  "Low worry about infection",
```


"Low risk perception",
"Endorsing conspiracy theories",
"Trust in alternative medicine",
"Internal health locus of control",
"Low sense of collective responsibility",
"Propensity to vote for L",
"Propensity to vote for 5SM",
"Propensity to vote for BOI",
"Distrust of science",
"Religion, pray",
"Digital media diet",
"Sex, female",
"Age, young",
"Educational level, low",
"Living in a rural region",
"Economic insecurity",
"Vaccine hesitancy",
"Low compliance with preventive behaviors",
"Disapproval of Government",
"Distrust of institutions")

```
shapes <- c(
  "circle",
  "circle",
  "circle",
  "circle",
  "circle",
  "circle",
  "circle",
  "circle",
  "circle",
  "circle",
  "circle",
  "circle",
  "circle",
  "square",
  "circle",
  "circle",
  "circle")
```

```
# Community detection function (modified to allow for iterations)
communityStability <- function(data, type, level, iterations)
{
  communityMemberships <- list()

  for (i in 1:iterations)
  {
```

```

fitGraph <- mgm(data, type, level, k = 2, verbatim = TRUE)
iGraph<- graph_from_adjacency_matrix(abs
                                     (fitGraph$pairwise$wadj), "undire
cted", weighted =
                                     TRUE)
  communityMemberships[[i]] <- cluster_walktrap(iGraph)$membership
}

communityOverlap <- matrix(NA, length(data), length(data))

for(j in 1:length(data))
{
  overlapPerNode <- matrix(NA, iterations, ncol(communityOverlap))

  for(k in 1: iterations)
  {
    overlapPerNode[k,] <- as.numeric(communityMemberships[[k]][j] == com
munityMemberships[[k]])
  }

  communityOverlap[j,] <- apply(overlapPerNode, 2, mean)
}

diag(communityOverlap) <- 0

return(list(communityMemberships = communityMemberships, communityOverla
p = communityOverlap))
}

```

Fit model

```

set.seed(1)
FitW3 <- mgm(W3, type, level, k = 2, binarySign = TRUE)

##      |
|                                             |
0%      |
|---|                                         |
4%      |
|-----|                                     |
9%      |
|-----|                                     |
13%     |
|-----|                                     |
17%     |
|-----|                                     |
22%     |
|-----|                                     |
26%     |
|-----|                                     |
30%     |
|-----|                                     |
35%     |

```

```

|-----|
39% |
|-----|
43% |
|-----|
48% |
|-----|
52% |
|-----|
57% |
|-----|
61% |
|-----|
65% |
|-----|
70% |
|-----|
74% |
|-----|
78% |
|-----|
83% |
|-----|
87% |
|-----|
91% |
|-----|
96% |
|-----| 1
00%
## Note that the sign of parameter estimates is stored separately; see ?mg
m

```

Community stability (Supplement S1)

```

#LOAD THE OBJECT down below TO SAVE TIME:
#CommunityStabTotal<-communityStability(W3, type, level, it=1000)

#Load the object instead:
CommunityStabTotal = readRDS(here("Input", "CommunityStabTotal.rds"))

# Visualize community stability
pdf(file = '../Output/Supplement/community_stability.pdf',paper = "USr",
    height = 9, width = 12)
qgraph(CommunityStabTotal$communityOverlap, layout = "spring",
    theme = "Borkulo", labels = shortnames,
    nodeNames = longnames,vsize=4.0,
    edge.labels=FALSE, legend = TRUE, legend.cex = 0.3)
dev.off()

## quartz_off_screen
## 2

# Vizualise detected communities
inputCommDetection_total<-CommunityStabTotal$communityOverlap

```

```

inputCommDetection_total[which(inputCommDetection_total <= .90)] <- 0

pdf(file = '../Output/Supplement/detected_communities.pdf', paper = "USr",
    height = 9, width = 12)
qgraph(inputCommDetection_total, layout = "spring", theme = "Borkulo",
    labels = shortnames, nodeNames = longnames, vsize=4.0,
    edge.labels=FALSE, legend = TRUE, legend.cex = 0.3)
dev.off()

## quartz_off_screen
##                2

Totalgroup_comm <- list(
  " "=c(1,2,12,20),
  " "=c(3,4),
  " "=c(5,8),
  " "=c(6,7,21),
  " "=c(9,11),
  " "=c(10,22,23),
  " "=c(13:19))

# define nice colors
Totalgroup_cols <- c("#BDE6F1", "#E96479", "#3FA796", "#B4CFB0",
                     "#DF7861", "#C3ACD0", "#FFD56F")

```

Plot no minimum (Supplement S1)

```

#enables theme colorblind because we don't need to specify edge.color
inputGraphMGM <- FitW3$pairwise$wadj
signsGraphMGM <- FitW3$pairwise$signs
signsGraphMGM[which(is.na(signsGraphMGM))] <- 1
inputGraphMGM <- inputGraphMGM*signsGraphMGM

# Plot
set.seed(1)
pdf(file = '../Output/Supplement/MGM_nomin.pdf', paper = "USr", height = 9,
    width = 12)
GraphMGM<-qgraph(inputGraphMGM,
  layout = "spring", theme = "Borkulo",
  labels = shortnames, nodeNames = longnames,
  cut = 0.10, maximum = 1,
  details = FALSE, vsize=6.0, shape = shapes,
  groups=Totalgroup_comm, color= Totalgroup_cols,
  legend = TRUE, legend.cex = 0.4, borders = FALSE)
dev.off()

## quartz_off_screen
##                2

```

Plot with predictability (Article)

```

#Predicted values
p_obj <- predict(FitW3, W3, errorCat = c("CC","nCC","CCmarg"), errorCon =
c("R2"))

#insert them in a list, and compute different metrics for categorical var

```

tables

```
error_list <- list()
error_list[[1]] = p_obj$errors[1,2]
error_list[[2]] = p_obj$errors[2,2]
  beyondmarg <- p_obj$errors[3,3]-p_obj$errors[3,5]
  error_list[[3]] = c(p_obj$errors[3,5],beyondmarg)
error_list[[4]] = p_obj$errors[4,2]
  beyondmarg <- p_obj$errors[5,3]-p_obj$errors[5,5]
  error_list[[5]] = c(p_obj$errors[5,5],beyondmarg)
error_list[[6]] = p_obj$errors[6,2]
error_list[[7]] = p_obj$errors[7,2]
error_list[[8]] = p_obj$errors[8,2]
error_list[[9]] = p_obj$errors[9,2]
error_list[[10]] = p_obj$errors[10,2]
error_list[[11]] = p_obj$errors[11,2]
error_list[[12]] = p_obj$errors[12,2]
  beyondmarg <- p_obj$errors[13,3]-p_obj$errors[13,5]
  error_list[[13]] = c(p_obj$errors[13,5],beyondmarg)
  beyondmarg <- p_obj$errors[14,3]-p_obj$errors[14,5]
  error_list[[14]] = c(p_obj$errors[14,5],beyondmarg)
  beyondmarg <- p_obj$errors[15,3]-p_obj$errors[15,5]
  error_list[[15]] = c(p_obj$errors[15,5],beyondmarg)
error_list[[16]] = p_obj$errors[16,2]
  beyondmarg <- p_obj$errors[17,3]-p_obj$errors[17,5]
  error_list[[17]] = c(p_obj$errors[17,5],beyondmarg)
  beyondmarg <- p_obj$errors[18,3]-p_obj$errors[18,5]
  error_list[[18]] = c(p_obj$errors[18,5],beyondmarg)
error_list[[19]] = p_obj$errors[19,2]
  beyondmarg <- p_obj$errors[20,3]-p_obj$errors[20,5]
  error_list[[20]] = c(p_obj$errors[20,5],beyondmarg)
error_list[[21]] = p_obj$errors[21,2]
error_list[[22]] = p_obj$errors[22,2]
error_list[[23]] = p_obj$errors[23,2]
```

List for Colors

```
color_list <- list()
color_list[[1]] = "#90B4D4"
color_list[[2]] = "#90B4D4"
color_list[[3]] = c("#ffa500", "#ff4300")
color_list[[4]] = "#90B4D4"
color_list[[5]] = c("#ffa500", "#ff4300")
color_list[[6]] = "#90B4D4"
color_list[[7]] = "#90B4D4"
color_list[[8]] = "#90B4D4"
color_list[[9]] = "#90B4D4"
color_list[[10]] = "#90B4D4"
color_list[[11]] = "#90B4D4"
color_list[[12]] = "#90B4D4"
color_list[[13]] = c("#ffa500", "#ff4300")
color_list[[14]] = c("#ffa500", "#ff4300")
color_list[[15]] = c("#ffa500", "#ff4300")
color_list[[16]] = "#90B4D4"
color_list[[17]] = c("#ffa500", "#ff4300")
color_list[[18]] = c("#ffa500", "#ff4300")
```

```

color_list[[19]] = "#90B4D4"
color_list[[20]] = c("#ffa500", "#ff4300")
color_list[[21]] = "#90B4D4"
color_list[[22]] = "#90B4D4"
color_list[[23]] = "#90B4D4"

# The final plot
set.seed(100)
GraphMGM<-qgraph(inputGraphMGM, pie = error_list, pieColor = color_list,
  layout = "spring", theme = "Borkulo",
  labels = shortnames, nodeNames = longnames,
  cut = 0.10, minimum = 0.06, maximum = 1,
  details = FALSE, vsize=6.0, shape = shapes,
  groups=Totalgroup_comm, color= Totalgroup_cols,
  legend = TRUE, legend.cex = 0.35, borders = TRUE,
  filetype="jpg", filename=here("Output", "Article", "MGM_min_06_predictability"))

```

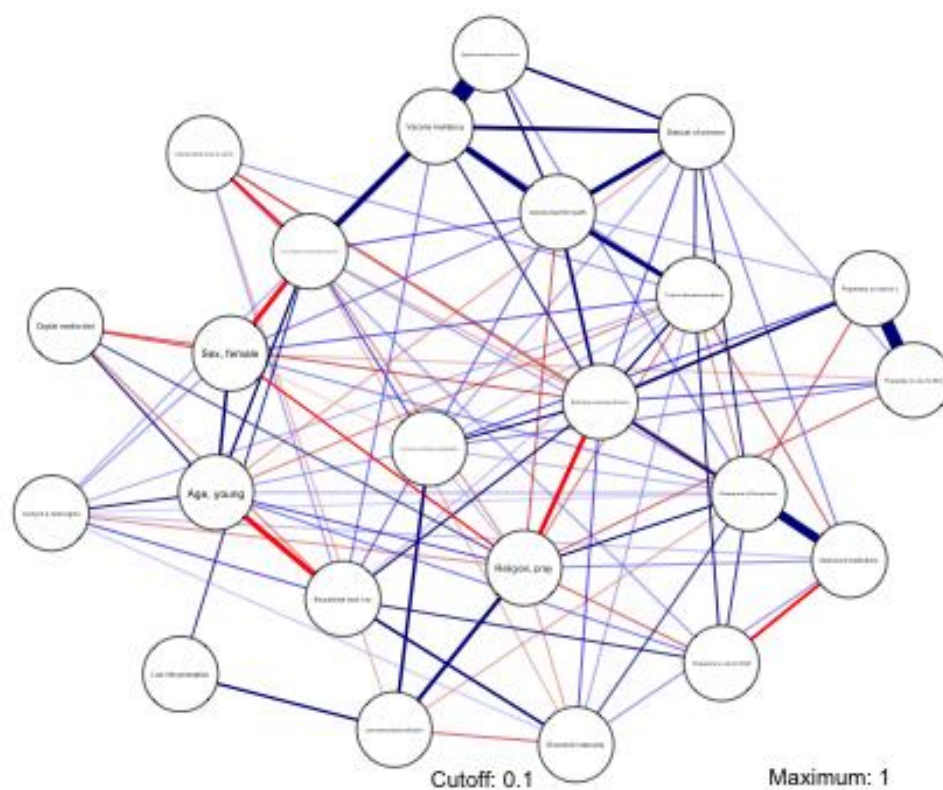
Output stored in /Users/art/Documents/Github/lavori in corso/papers/vaccine/Processing//Users/art/Documents/Github/lavori in corso/papers/vaccine/Output/Article/MGM_min_06_predictability.jpg

Centrality estimates

```

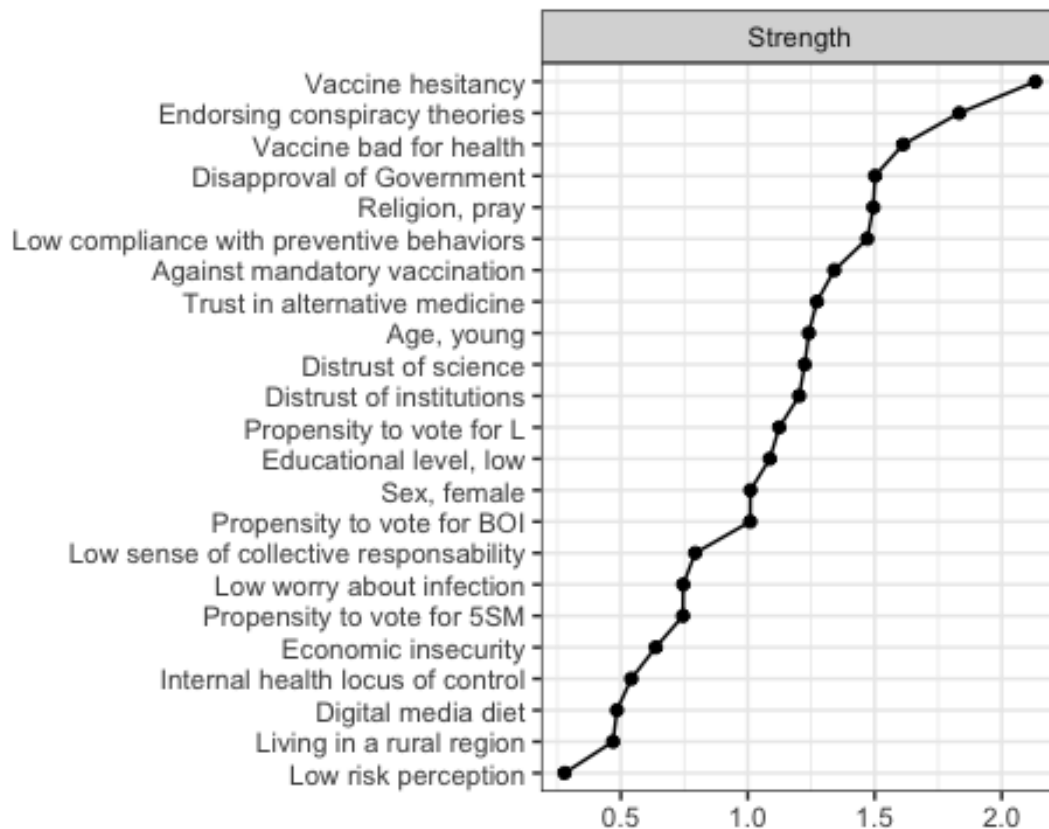
#graph with complete labels for centrality plot
GraphMGMCENT<-qgraph(inputGraphMGM,
  layout = "spring", theme = "colorblind",
  labels = longnames, minimum = 0,
  cut = 0.10, maximum = 1, details = TRUE,
  legend = FALSE)

```



```
#centrality Total
centMGM <- centralityTable(GraphMGMCENT, standardized = FALSE, signed = FALSE)

#CentralityPlot Total
centrality_table = centralityPlot(GraphMGMCENT, include = c("Strength"), scale = "raw",
                                orderBy = "Strength", signed = FALSE) + theme_nice() + theme(
axis.text.y=element_text(hjust=1))
```



```
ggsave(here("Output", "Article", "Centrality_Table.jpg"), centrality_table
)
```

```
## Saving 5 x 4 in image
```

Degree centrality

```
#Convert to igraph
```

```
igraph_graph <- graph_adjacency(FitW3$pairwise$wadj, mode="undirected", we
ighted = TRUE)
```

```
#Degree centrality
```

```
degree_cent = degree(igraph_graph)
```

```
#Merge vectors
```

```
degree_data <- data.frame(names(W3), degree_cent)
```

ASPL

```
#Shortest Path Length with tnet from opshal
```

```
spl_o = distance_w(FitW3$pairwise$wadj, directed=NULL, subsample=1)
```

```
spl_o <- spl_o[upper.tri(spl_o)]
```

```
aspl_o <- mean(spl_o)
```

ASPL = 1.563595

Bootstrap (Supplement S1)

Edge accuracy

```
#edge weight accuracy: non parametric bootstrap with 8 cores
#edgeacc = bootnet(W3, nBoots = 1000, nCores = 8, 'mgm')

#Load the object instead:
edgeacc = readRDS(here("Input", "edgeacc.rds"))

#plot 1
pdf('../Output/Supplement/robustness/edge_accuracy.pdf', height = 70, width = 50)
plot(edgeacc, labels = longnames, order = "sample")
dev.off()

## quartz_off_screen
##                2

#Plot 2
pdf('../Output/Supplement/robustness/edge_accuracy_CI.pdf', height = 70, width = 50)
plot(edgeacc, plot = "interval", split0 = TRUE, order="sample", labels=longnames)
dev.off()

## quartz_off_screen
##                2

#summary
saummary_edgeacc = summary(edgeacc, statistics = c("edge", "strength"),
                           perNode = FALSE, rank = FALSE)
```

Centrality stability (Supplement S1)

```
# case dropping bootstrap
#centstab = bootnet(W3, nBoots = 1000, 'mgm', type = "case", nCores = 8)

#Load the object instead:
centstab = readRDS(here("Input", "centstab.rds"))

#plot 1
pdf('../Output/Supplement/robustness/Centrality_stability.pdf', height = 70, width = 50)
plot(centstab, "Strength", perNode = TRUE, labels = longnames,
     subsetRange = c(100,50))
dev.off()

## quartz_off_screen
##                2

#Plot 2
pdf('../Output/Supplement/robustness/Centrality_stability_CI.pdf', height = 70, width = 50)
plot(centstab, "Strength", CIstyle = "quantiles")
dev.off()
```

```

## quartz_off_screen
##          2

#CS-coefficient (result should be above 0.25, better if above 0.5)
constab = corStability(censtab)

## === Correlation Stability Analysis ===
##
## Sampling levels tested:
##   nPerson Drop%   n
## 1      385  75.0  86
## 2      505  67.2 111
## 3      625  59.4  87
## 4      744  51.7  99
## 5      864  43.9 100
## 6      984  36.1  99
## 7     1104  28.3  97
## 8     1223  20.6 105
## 9     1343  12.8  94
## 10    1463   5.0 122
##
## Maximum drop proportions to retain correlation of 0.7 in at least 95% o
f the samples:
##
## edge: 0.75 (CS-coefficient is highest level tested)
##   - For more accuracy, run bootnet(..., caseMin = 0.672, caseMax = 1)
##
## strength: 0.75 (CS-coefficient is highest level tested)
##   - For more accuracy, run bootnet(..., caseMin = 0.672, caseMax = 1)
##
## Accuracy can also be increased by increasing both 'nBoots' and 'caseN'.

```

Testing Edge and centrality differences (Supplement S1)

```

# Test: difference of weight ties 2-3 vs 4-5
differenceTest(edgeacc, 2--3, 3--4, "strength")

## Expected significance level given number of bootstrap samples is approx
imately: 0.05

##           id1      id2  measure      lower      upper significant
## 1 conspiracy h_locus strength -1.400756 -0.5711839          TRUE

# Plot test results for every edge weight in the network
pdf('../Output/Supplement/robustness/test_edges.pdf', height = 70, width =
50)
plot(edgeacc, "edge", plot = "difference", onlyNonZero = TRUE, order = "sa
mple",
      labels = T)

## Expected significance level given number of bootstrap samples is approx
imately: 0.05

dev.off()

## quartz_off_screen
##          2

```

```

# Test: difference of strength of node 5 [conspiracy] vs 20 [hesitancy] (if the bootstrapped CI include 0, they do not differ)
test_conspiracy_vs_hesitancy = differenceTest(edgeacc, 5, 20, "strength")

## Expected significance level given number of bootstrap samples is approximately: 0.05

#test_conspiracy_vs_distrust_gov = differenceTest(edgeacc, 5, 22, "strength")
#test_conspiracy_vs_vac_bad = differenceTest(edgeacc, 5, 1, "strength")
#test_conspiracy_vs_pray = differenceTest(edgeacc, 5, 13, "strength")

# Plot test results for every edge weight in the network
pdf('../Output/Supplement/robustness/test_strengths.pdf', height = 70, width = 50)
plot(edgeacc, "strength", order = "mean", labels = T)

## Expected significance level given number of bootstrap samples is approximately: 0.05

dev.off()

## quartz_off_screen
##                2

```

Additional robustness check (Supplement S1)

```

#Creating serial number for each row
W3_robustness = W3
W3_robustness$id = 1:nrow(W3)

#Splitting randomly the original data frame in two partitions
W3_robustness_a = W3_robustness %>% dplyr::sample_frac(.50)
W3_robustness_b = dplyr::anti_join(W3_robustness, W3_robustness_a, by = 'id')

#Remove id variable
W3_robustness_a = W3_robustness_a %>% dplyr::select(-id)
W3_robustness_b = W3_robustness_b %>% dplyr::select(-id)

#Fitting two mgm
set.seed(1)
FitW3_robustness_a <- mgm(W3_robustness_a, type, level, k = 2, binarySign = TRUE)

##      |
|      |
0%      |
|---    |
4%      |
|-----|
9%      |
|-----|
13%     |
|-----|
17%     |

```

```

|-----|
22% |
|-----|
26% |
|-----|
30% |
|-----|
35% |
|-----|
39% |
|-----|
43% |
|-----|
48% |
|-----|
52% |
|-----|
57% |
|-----|
61% |
|-----|
65% |
|-----|
70% |
|-----|
74% |
|-----|
78% |
|-----|
83% |
|-----|
87% |
|-----|
91% |
|-----|
96% |
|-----| 1
00%
## Note that the sign of parameter estimates is stored separately; see ?mg
m

set.seed(1)
FitW3_robustness_b <- mgm(W3_robustness_b, type, level, k = 2, binarySign
= TRUE)

## |
|
0% |
|---|
4% |
|-----|
9% |
|-----|
13% |

```

```

|-----|
17% |
|-----|
22% |
|-----|
26% |
|-----|
30% |
|-----|
35% |
|-----|
39% |
|-----|
43% |
|-----|
48% |
|-----|
52% |
|-----|
57% |
|-----|
61% |
|-----|
65% |
|-----|
70% |
|-----|
74% |
|-----|
78% |
|-----|
83% |
|-----|
87% |
|-----|
91% |
|-----|
96% |
|-----| 1
00%

```

```

## Note that the sign of parameter estimates is stored separately; see ?mg
m

```

```

#Network Comparison Test between the two mgm

```

```

set.seed(1)

```

```

robustness_a_b = NCT(W3_robustness_a, W3_robustness_b, it = 1000,
  test.edges=TRUE, edges="all",
  progressbar=FALSE, p.adjust.methods= c("bonferroni"),
  test.centraliity=TRUE,
  centraliity=c("strength"),nodes="all")

```

```

#Visualize results of edge tests

```

```

inputNCTgraph <- FitW3_robustness_a$pairwise$wadj - FitW3_robustness_b$pai
rwise$wadj

```

```

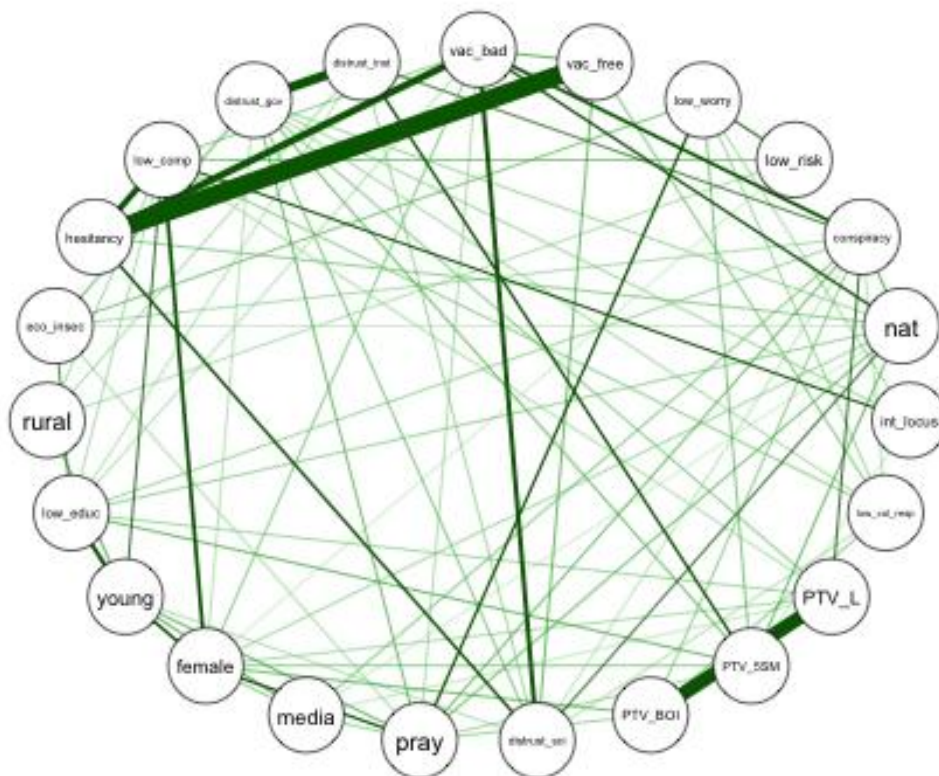
inputNCTgraph[upper.tri(inputNCTgraph)][which(robustness_a_b$inv.pvals`p
-value` >= .05)] <- 0
inputNCTgraph <- forceSymmetric(inputNCTgraph)

pdf(here("Output", "Supplement", "nct.pdf"))
nct_plot = qgraph(inputNCTgraph, labels = shortnames, edge.labels = TRUE)
dev.off()

## quartz_off_screen
##                2

#Visualize results of centrality tests
#Calculate strength
Graph_robustness_a <- qgraph(FitW3_robustness_a$pairwise$wadj, labels = sho
rtnames)

```

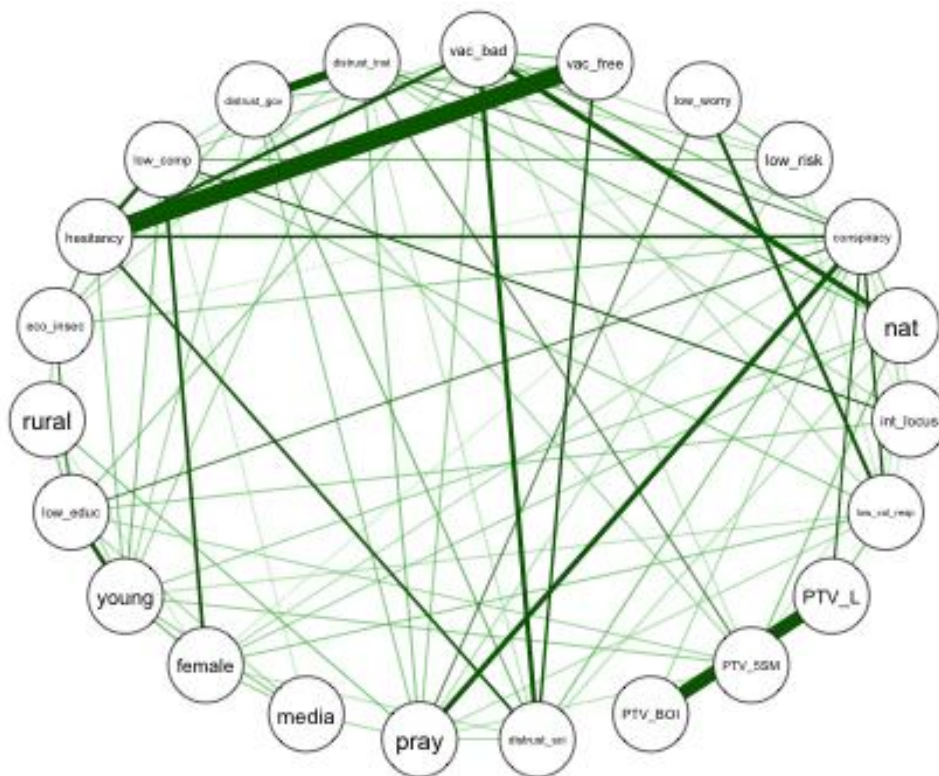


```

cent_robustness_a <- centralityTable(Graph_robustness_a)

Graph_robustness_b <- qgraph(FitW3_robustness_b$pairwise$wadj, labels = sho
rtnames)

```



```
cent_robustness_b <- centralityTable(Graph_robustness_b)
```

#isolate what is statistically different

`robustness_a_b$diffcen.pval` *#no differences are statistically significant*

##	strength
## vac_bad	1.000
## vac_ob	1.000
## worry	1.000
## risk	1.000
## conspiracy	1.000
## nat	1.000
## h_locus	1.000
## he_eco	1.000
## PTV_L	1.000
## PTV_5SM	1.000
## PTV_BOI	1.000
## tr_sci	1.000
## pray	1.000
## media	1.000
## sex	1.000
## age	0.046
## educ	1.000
## reg	0.184
## eco_insec	1.000
## hesitancy	1.000
## low_comp	1.000

```
## distrust_gov      1.000
## distrust_inst     1.000

robustness_a_b$diffcen.real #raw differences in centrality

##                strength
## vac_bad         0.133839956
## vac_ob          0.008350283
## worry           0.098534698
## risk            0.146727652
## conspiracy       -0.026317984
## nat             0.184549079
## h_locus         0.148037506
## he_eco          -0.094034582
## PTV_L           -0.027806009
## PTV_5SM         0.176776702
## PTV_BOI         0.140999444
## tr_sci          0.086101789
## pray           0.258077991
## media           0.189061865
## sex             0.097210152
## age             0.420278912
## educ            0.168005877
## reg             0.123570893
## eco_insec       0.161828352
## hesitancy       0.092382804
## low_comp        0.191480095
## distrust_gov    0.004318075
## distrust_inst   0.162524112
```

Backbone (Article)

#model selection

```
backbone.suggest(FitW3$pairwise$wadj)
```

The disparity filter is suggested. Type "?disparity" for more information.

application of the disparity filter

```
backbone_matrix <- disparity(
  inputGraphMGM,
  alpha = 0.05,
  signed = FALSE,
  mtc = "none",
  class = "original",
  narrative = TRUE)
```

##

=== Suggested manuscript text and citations ===

We used the backbone package for R (v2.1.2; Neal, 2022) to extract the unweighted backbone of a weighted and undirected unipartite network containing 23 nodes. An edge was retained in the backbone if its weight was statistically significant ($\alpha = 0.05$) using the disparity filter (Serrano e

t al., 2009). This reduced the number of edges by 82%, and reduced the number of connected nodes by 26.1%.

##

Neal, Z. P. 2022. backbone: An R Package to Extract Network Backbones. PLOS ONE, 17, e0269137. <https://doi.org/10.1371/journal.pone.0269137>

##

Serrano, M. A., Boguna, M., & Vespignani, A. (2009). Extracting the multiscale backbone of complex weighted networks. Proceedings of the National Academy of Sciences, 106(16), 6483-6488. <https://doi.org/10.1073/pnas.0808904106>

#plot

set.seed(100)

```
GraphMGM<-qgraph(backbone_matrix,
  layout = "spring", theme = "Borkulo",
  labels = shortnames,nodeNames = longnames,
  details = FALSE, vsize=6.0, shape = shapes,
  groups=Totalgroup_comm, color= Totalgroup_cols,
  legend = TRUE, legend.cex = 0.35, borders = FALSE,
  filetype="jpg", filename=here("Output", "Article", "backbone.jpg"))
```

Output stored in /Users/art/Documents/Github/lavori in corso/papers/vaccine/Processing//Users/art/Documents/Github/lavori in corso/papers/vaccine/Output/Article/backbone.jpg.jpg

Regressions (Supplement S1)

#Fitting logistic regressions

```
model_vac_bad <- glm( hesitancy ~ vac_bad + sex + age + educ + reg + eco_insec,
```

```
                      data = W3, family = binomial)
```

```
model_vac_ob <- glm( hesitancy ~ vac_ob + sex + age + educ + reg + eco_insec,
```

```
                      data = W3, family = binomial)
```

```
model_worry <- glm( hesitancy ~ worry + sex + age + educ + reg + eco_insec,
```

```
                      data = W3, family = binomial)
```

```
model_risk <- glm( hesitancy ~ risk + sex + age + educ + reg + eco_insec,
```

```
                      data = W3, family = binomial)
```

```
model_conspiracy <- glm( hesitancy ~ conspiracy + sex + age + educ + reg + eco_insec,
```

```
                      data = W3, family = binomial)
```

```
model_nat <- glm( hesitancy ~ nat + sex + age + educ + reg + eco_insec,
```

```
                      data = W3, family = binomial)
```

```
model_h_locus <- glm( hesitancy ~ h_locus + sex + age + educ + reg + eco_insec,
```

```

        data = W3, family = binomial)

model_he_eco <- glm( hesitancy ~ he_eco + sex + age + educ + reg + eco_insec,
        data = W3, family = binomial)

model_PTV_L <- glm( hesitancy ~ PTV_L + sex + age + educ + reg + eco_insec,
        data = W3, family = binomial)

model_PTV_5SM <- glm( hesitancy ~ PTV_5SM + sex + age + educ + reg + eco_insec,
        data = W3, family = binomial)

model_PTV_BOI <- glm( hesitancy ~ PTV_BOI + sex + age + educ + reg + eco_insec,
        data = W3, family = binomial)

model_tr_sci <- glm( hesitancy ~ tr_sci + sex + age + educ + reg + eco_insec,
        data = W3, family = binomial)

model_pray <- glm( hesitancy ~ pray + sex + age + educ + reg + eco_insec,
        data = W3, family = binomial)

model_media <- glm( hesitancy ~ media + sex + age + educ + reg + eco_insec,
        data = W3, family = binomial)

model_low_comp <- glm( hesitancy ~ low_comp + sex + age + educ + reg + eco_insec,
        data = W3, family = binomial)

model_distrust_gov <- glm( hesitancy ~ distrust_gov + sex + age + educ + reg + eco_insec,
        data = W3, family = binomial)

model_distrust_inst <- glm( hesitancy ~ distrust_inst + sex + age + educ + reg + eco_insec,
        data = W3, family = binomial)

#Additional model for final table
model_educ <- glm( hesitancy ~ educ + sex + age + reg + eco_insec,
        data = W3, family = binomial)

#coeff
coeff_educ = exp(extract_numeric(model_educ$coefficients[2]))

## extract_numeric() is deprecated: please use readr::parse_number() instead

#significance
summary(model_educ)$coeff[-1,4] < 0.05

```

##	educ	sex	age	reg	eco_insec
##	TRUE	TRUE	FALSE	FALSE	TRUE

Export regression table for additional material (Supplement S1)

```
#model_vac_bad
tab_model(list(model_vac_bad),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("vac_bad"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
  auto.label = FALSE,
  file = here("Output", "Supplement", "regressions", "model_vac_ba
d.doc")
)
```

Logit regression model on vaccine hesitancy

vac_bad

Odds Ratios

(Intercept)

0.00 *** (0.00 – 0.01)

vac_bad

3.39 *** (2.95 – 3.93)

sex

1.04 (0.75 – 1.44)

age

1.01 ** (1.00 – 1.02)

educ

1.28 (0.90 – 1.83)

reg

1.12 (0.80 – 1.55)

eco_insec

1.06 (0.82 – 1.35)

Observations

1540

R2 Tjur

0.319

AIC

986.178

- p<0.1 ** p<0.05 *** p<0.01

```
#model_vac_ob
tab_model(list(model_vac_ob),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("vac_ob"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
  auto.label = FALSE,
  file = here("Output", "Supplement", "regressions", "model_vac_ob
.doc")
)
```

Logit regression model on vaccine hesitancy

vac_ob

Odds Ratios

(Intercept)

0.00 *** (0.00 – 0.00)

vac_ob

5.67 *** (4.71 – 6.94)

sex

0.98 (0.67 – 1.43)

age

1.01 * (1.00 – 1.03)

educ

1.61 ** (1.07 – 2.43)

reg

1.09 (0.74 – 1.60)

eco_insec

0.79 (0.58 – 1.07)

Observations

1540

R2 Tjur

0.509

AIC

736.441

- p<0.1 ** p<0.05 *** p<0.01

```
#model_worry
tab_model(list(model_worry),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("worry"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
  auto.label = FALSE,
  file = here("Output", "Supplement", "regressions", "model_worry.
doc")
)
```

Logit regression model on vaccine hesitancy

worry

Odds Ratios

(Intercept)

0.05 *** (0.02 – 0.11)

worry

1.12 (0.84 – 1.50)

sex

1.41 ** (1.08 – 1.85)

age

1.00 (0.99 – 1.01)

educ

1.77 ***(1.32 – 2.39)

reg

1.07 (0.82 – 1.40)

eco_insec

1.26 **(1.02 – 1.56)

Observations

1540

R2 Tjur

0.021

AIC

1394.191

- p<0.1 ** p<0.05 *** p<0.01

#model_risk

```
tab_model(list(model_risk),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("risk"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
  auto.label = FALSE,
  file = here("Output", "Supplement", "regressions", "model_risk.doc")
)
```

Logit regression model on vaccine hesitancy

risk

Odds Ratios

(Intercept)

0.02 ***(0.01 – 0.06)

risk

1.31 ***(1.10 – 1.57)

sex

1.40 **(1.07 – 1.84)

age

1.00 (0.99 – 1.01)

educ

1.76 *** (1.31 – 2.38)

reg

1.05 (0.80 – 1.38)

eco_insec

1.28 ** (1.04 – 1.58)

Observations

1540

R2 Tjur

0.027

AIC

1385.843

- p<0.1 ** p<0.05 *** p<0.01

```
#model_conspiracy
tab_model(list(model_conspiracy),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("conspiracy"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
  auto.label = FALSE,
  file = here("Output", "Supplement", "regressions", "model_conspi
racy.doc")
)
```

Logit regression model on vaccine hesitancy

conspiracy

Odds Ratios

(Intercept)

0.03 *** (0.01 – 0.07)

conspiracy

5.18 *** (3.79 – 7.17)

sex

1.31 * (0.99 – 1.74)

age

1.01 (1.00 – 1.02)

educ

1.44 ** (1.06 – 1.96)

reg

1.05 (0.79 – 1.39)

eco_insec

1.12 (0.90 – 1.38)

Observations

1540

R2 Tjur

0.095

AIC

1273.608

- p<0.1 ** p<0.05 *** p<0.01

#model_nat

```
tab_model(list(model_nat),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("nat"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
  auto.label = FALSE,
  file = here("Output", "Supplement", "regressions", "model_nat.do
c")
)
```

Logit regression model on vaccine hesitancy

nat

Odds Ratios

(Intercept)

0.02 *** (0.01 – 0.04)

nat

1.32 *** (1.26 – 1.39)

sex

1.17 (0.88 – 1.55)

age

1.01 ** (1.00 – 1.02)

educ

1.48 ** (1.09 – 2.03)

reg

1.00 (0.75 – 1.33)

eco_insec

1.18 (0.94 – 1.47)

Observations

1540

R2 Tjur

0.119

AIC

1251.696

- p<0.1 ** p<0.05 *** p<0.01

```
#model_h_locus
tab_model(list(model_h_locus),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("h_locus"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
  auto.label = FALSE,
  file = here("Output", "Supplement", "regressions", "model_h_locu
s.doc")
)
```

Logit regression model on vaccine hesitancy

h_locus

Odds Ratios

(Intercept)

0.29 *** (0.13 – 0.65)

h_locus

0.78 *** (0.74 – 0.82)

sex

1.47 *** (1.11 – 1.95)

age

1.00 (0.99 – 1.01)

educ

1.92 *** (1.41 – 2.62)

reg

1.07 (0.81 – 1.42)

eco_insec

1.17 (0.94 – 1.45)

Observations

1540

R2 Tjur

0.095

AIC

1295.937

- p<0.1 ** p<0.05 *** p<0.01

#model_he_eco

```
tab_model(list(model_he_eco),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("he_eco"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
```

```

auto.label = FALSE,
file = here("Output", "Supplement", "regressions", "model_he_eco
.doc")
)

```

Logit regression model on vaccine hesitancy

he_eco

Odds Ratios

(Intercept)

0.02 *** (0.01 – 0.05)

he_eco

1.24 *** (1.17 – 1.31)

sex

1.49 *** (1.13 – 1.96)

age

1.00 (0.99 – 1.01)

educ

1.67 *** (1.23 – 2.26)

reg

1.06 (0.80 – 1.39)

eco_insec

1.22 * (0.98 – 1.50)

Observations

1540

R2 Tjur

0.065

AIC

1332.312

- p<0.1 ** p<0.05 *** p<0.01

```

#model_PTV_L
tab_model(list(model_PTV_L),
  show.p = TRUE,
  p.style = "stars",

```

```

p.threshold = c(0.1, 0.05, 0.01),
dv.labels = c("PTV_L"),
collapse.ci = TRUE,
show.aic = TRUE,
title = "Logit regression model on vaccine hesitancy",
string.pred = " ",
auto.label = FALSE,
file = here("Output", "Supplement", "regressions", "model_PTV_L.
doc")
)

```

Logit regression model on vaccine hesitancy

PTV_L

Odds Ratios

(Intercept)

0.04 ***(0.02 – 0.08)

PTV_L

1.13 ***(1.09 – 1.17)

sex

1.37 *(1.04 – 1.80)

age

1.01 (1.00 – 1.02)

educ

1.60 ***(1.19 – 2.17)

reg

1.02 (0.77 – 1.35)

eco_insec

1.29 *(1.05 – 1.60)

Observations

1540

R2 Tjur

0.047

AIC

1350.314

- p<0.1 ** p<0.05 *** p<0.01

```
#model_PTV_5SM
tab_model(list(model_PTV_5SM),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("PTV_5SM"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
  auto.label = FALSE,
  file = here("Output", "Supplement", "regressions", "model_PTV_5S
M.doc")
)
```

Logit regression model on vaccine hesitancy

PTV_5SM

Odds Ratios

(Intercept)

0.06 *** (0.03 – 0.12)

PTV_5SM

0.98 (0.94 – 1.02)

sex

1.41 ** (1.08 – 1.84)

age

1.00 (0.99 – 1.01)

educ

1.79 *** (1.33 – 2.42)

reg

1.07 (0.82 – 1.41)

eco_insec

1.26 ** (1.02 – 1.55)

Observations

1540

R2 Tjur

0.021

AIC

1393.970

- p<0.1 ** p<0.05 *** p<0.01

```
#model_PTV_BOI
tab_model(list(model_PTV_BOI),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("PTV_BOI"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
  auto.label = FALSE,
  file = here("Output", "Supplement", "regressions", "model_PTV_BOI.doc")
)
```

Logit regression model on vaccine hesitancy

PTV_BOI

Odds Ratios

(Intercept)

0.03 *** (0.02 – 0.07)

PTV_BOI

1.15 *** (1.11 – 1.19)

sex

1.43 ** (1.09 – 1.89)

age

1.01 (1.00 – 1.02)

educ

1.59 *** (1.18 – 2.16)

reg

1.03 (0.78 – 1.36)

eco_insec

1.29 ** (1.05 – 1.60)

Observations

1540

R2 Tjur

0.054

AIC

1340.696

- p<0.1 ** p<0.05 *** p<0.01

```
#model_tr_sci
tab_model(list(model_tr_sci),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("tr_sci"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
  auto.label = FALSE,
  file = here("Output", "Supplement", "regressions", "model_tr_sci
.doc")
)
```

Logit regression model on vaccine hesitancy

tr_sci

Odds Ratios

(Intercept)

0.00 *** (0.00 – 0.01)

tr_sci

3.37 *** (2.93 – 3.91)

sex

1.22 (0.89 – 1.68)

age

1.01 ** (1.00 – 1.03)

educ

1.58 ** (1.11 – 2.25)

reg

0.96 (0.69 – 1.32)

eco_insec

1.04 (0.81 – 1.34)

Observations

1540

R2 Tjur

0.300

AIC

1015.488

- p<0.1 ** p<0.05 *** p<0.01

```
#model_pray
tab_model(list(model_pray),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("pray"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
  auto.label = FALSE,
  file = here("Output", "Supplement", "regressions", "model_pray.d
oc")
)
```

Logit regression model on vaccine hesitancy

pray

Odds Ratios

(Intercept)

0.06 *** (0.03 – 0.13)

pray

0.79 * (0.60 – 1.04)

sex

1.37 ** (1.05 – 1.80)

age

1.00 (1.00 – 1.01)

educ

1.75 ***(1.30 – 2.37)

reg

1.06 (0.81 – 1.40)

eco_insec

1.25 *(1.01 – 1.54)

Observations

1540

R2 Tjur

0.022

AIC

1391.908

- p<0.1 ** p<0.05 *** p<0.01

#model_media

```
tab_model(list(model_media),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("media"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
  auto.label = FALSE,
  file = here("Output", "Supplement", "regressions", "model_media.
doc")
)
```

Logit regression model on vaccine hesitancy

media

Odds Ratios

(Intercept)

0.06 ***(0.03 – 0.12)

media

0.99 (0.75 – 1.30)

sex

1.41 ******(1.07 – 1.84)

age

1.00 (0.99 – 1.01)

educ

1.76 *******(1.31 – 2.38)

reg

1.07 (0.82 – 1.41)

eco_insec

1.26 ******(1.02 – 1.55)

Observations

1540

R2 Tjur

0.020

AIC

1394.798

- p<0.1 ****** p<0.05 ******* p<0.01

#model_low_comp

```
tab_model(list(model_low_comp),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("low_comp"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
  auto.label = FALSE,
  file = here("Output", "Supplement", "regressions", "model_low_co
mp.doc")
)
```

Logit regression model on vaccine hesitancy

low_comp

Odds Ratios

(Intercept)

0.03 *******(0.02 – 0.07)

low_comp
1.57 *** (1.45 – 1.69)

sex
1.98 *** (1.47 – 2.69)

age
0.99 (0.98 – 1.00)

educ
1.81 *** (1.32 – 2.50)

reg
0.97 (0.72 – 1.30)

eco_insec
1.29 ** (1.03 – 1.61)

Observations
1540

R2 Tjur
0.152

AIC
1234.701

- p<0.1 ** p<0.05 *** p<0.01

```
#model_distrust_gov
tab_model(list(model_distrust_gov),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("distrust_gov"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
  auto.label = FALSE,
  file = here("Output", "Supplement", "regressions", "model_distrust_gov.doc")
)
```

Logit regression model on vaccine hesitancy

distrust_gov

Odds Ratios

(Intercept)

0.02 *** (0.01 – 0.04)

distrust_gov

1.43 *** (1.34 – 1.52)

sex

1.39 ** (1.04 – 1.84)

age

1.00 (0.99 – 1.01)

educ

1.78 *** (1.30 – 2.43)

reg

0.98 (0.73 – 1.31)

eco_insec

0.98 (0.78 – 1.22)

Observations

1540

R2 Tjur

0.125

AIC

1254.662

- p<0.1 ** p<0.05 *** p<0.01

```
#model_distrust_inst
tab_model(list(model_distrust_inst),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("distrust_inst"),
  collapse.ci = TRUE,
  show.aic = TRUE,
  title = "Logit regression model on vaccine hesitancy",
  string.pred = " ",
  auto.label = FALSE,
  file = here("Output", "Supplement", "regressions", "model_distrust_inst.doc")
)
```

Logit regression model on vaccine hesitancy

distrust_inst

Odds Ratios

(Intercept)

0.01 *** (0.01 – 0.03)

distrust_inst

1.38 *** (1.30 – 1.47)

sex

1.46 *** (1.10 – 1.93)

age

1.01 (1.00 – 1.02)

educ

1.66 *** (1.22 – 2.26)

reg

1.00 (0.75 – 1.32)

eco_insec

1.02 (0.82 – 1.27)

Observations

1540

R2 Tjur

0.112

AIC

1271.741

- p<0.1 ** p<0.05 *** p<0.01

```
#model_vac_bad
tab_model(list(model_vac_bad),
  show.p = TRUE,
  p.style = "stars",
  p.threshold = c(0.1, 0.05, 0.01),
  dv.labels = c("vac_bad"),
  collapse.ci = TRUE,
  show.aic = TRUE,
```

```

    title = "Logit regression model on vaccine hesitancy",
    string.pred = " ",
    auto.label = FALSE,
    file = here("Output", "Supplement", "regressions", "model_vac_ba
d.doc")
)

```

Logit regression model on vaccine hesitancy

vac_bad

Odds Ratios

(Intercept)

0.00 *** (0.00 – 0.01)

vac_bad

3.39 *** (2.95 – 3.93)

sex

1.04 (0.75 – 1.44)

age

1.01 ** (1.00 – 1.02)

educ

1.28 (0.90 – 1.83)

reg

1.12 (0.80 – 1.55)

eco_insec

1.06 (0.82 – 1.35)

Observations

1540

R2 Tjur

0.319

AIC

986.178

- p<0.1 ** p<0.05 *** p<0.01

```

#Additional model for final table
tab_model(list(model_educ),

```

```

    show.p = TRUE,
    p.style = "stars",
    p.threshold = c(0.1, 0.05, 0.01),
    dv.labels = c("educ"),
    collapse.ci = TRUE,
    show.aic = TRUE,
    title = "Logit regression model on vaccine hesitancy",
    string.pred = " ",
    auto.label = FALSE,
    file = here("Output", "Supplement", "regressions", "model_educ.d
oc")
)

```

Logit regression model on vaccine hesitancy

educ

Odds Ratios

(Intercept)

0.06 ***(0.03 – 0.12)

educ

1.76 ***(1.31 – 2.38)

sex

1.41 **(1.07 – 1.84)

age

1.00 (0.99 – 1.01)

reg

1.07 (0.82 – 1.41)

eco_insec

1.26 **(1.02 – 1.55)

Observations

1540

R2 Tjur

0.020

AIC

1392.803

- p<0.1 ** p<0.05 *** p<0.01

Scatterplot data

#Gathering coefficients

```
coeff = extract_numeric(model_vac_bad$coefficients[2])

## extract_numeric() is deprecated: please use readr::parse_number() instead

coeff[2] = extract_numeric(model_vac_ob$coefficients[2])

## extract_numeric() is deprecated: please use readr::parse_number() instead

coeff[3] = extract_numeric(model_worry$coefficients[2])

## extract_numeric() is deprecated: please use readr::parse_number() instead

coeff[4] = extract_numeric(model_risk$coefficients[2])

## extract_numeric() is deprecated: please use readr::parse_number() instead

coeff[5] = extract_numeric(model_conspiracy$coefficients[2])

## extract_numeric() is deprecated: please use readr::parse_number() instead

coeff[6] = extract_numeric(model_nat$coefficients[2])

## extract_numeric() is deprecated: please use readr::parse_number() instead

coeff[7] = extract_numeric(model_h_locus$coefficients[2])

## extract_numeric() is deprecated: please use readr::parse_number() instead

coeff[8] = extract_numeric(model_he_eco$coefficients[2])

## extract_numeric() is deprecated: please use readr::parse_number() instead

coeff[9] = extract_numeric(model_PTV_L$coefficients[2])

## extract_numeric() is deprecated: please use readr::parse_number() instead

coeff[10] = extract_numeric(model_PTV_5SM$coefficients[2])

## extract_numeric() is deprecated: please use readr::parse_number() instead

coeff[11] = extract_numeric(model_PTV_BOI$coefficients[2])

## extract_numeric() is deprecated: please use readr::parse_number() instead
```



```

coeff[12] = extract_numeric(model_tr_sci$coefficients[2])
## extract_numeric() is deprecated: please use readr::parse_number() instead

coeff[13] = extract_numeric(model_pray$coefficients[2])
## extract_numeric() is deprecated: please use readr::parse_number() instead

coeff[14] = extract_numeric(model_media$coefficients[2])
## extract_numeric() is deprecated: please use readr::parse_number() instead

coeff[15] = extract_numeric(model_low_comp$coefficients[2])
## extract_numeric() is deprecated: please use readr::parse_number() instead

coeff[16] = extract_numeric(model_distrust_gov$coefficients[2])
## extract_numeric() is deprecated: please use readr::parse_number() instead

coeff[17] = extract_numeric(model_distrust_inst$coefficients[2])
## extract_numeric() is deprecated: please use readr::parse_number() instead

  #Transform in odds ratio
  coeff_exp = unlist(lapply(coeff, exp))

#Significance
summary(model_vac_bad)$coeff[-1,4] < 0.05

##   vac_bad      sex      age      educ      reg eco_insec
##   TRUE      FALSE    TRUE    FALSE    FALSE    FALSE

summary(model_vac_ob)$coeff[-1,4] < 0.05

##   vac_ob      sex      age      educ      reg eco_insec
##   TRUE      FALSE    FALSE    TRUE    FALSE    FALSE

summary(model_worry)$coeff[-1,4] < 0.05 #not significant

##   worry      sex      age      educ      reg eco_insec
##   FALSE    TRUE    FALSE    TRUE    FALSE    TRUE

summary(model_risk)$coeff[-1,4] < 0.05

##   risk      sex      age      educ      reg eco_insec
##   TRUE    TRUE    FALSE    TRUE    FALSE    TRUE

summary(model_conspiracy)$coeff[-1,4] < 0.05

## conspiracy      sex      age      educ      reg eco_insec
##   TRUE      FALSE    FALSE    TRUE    FALSE    FALSE

```

```

summary(model_nat)$coeff[-1,4] < 0.05

##      nat      sex      age      educ      reg eco_insec
##      TRUE     FALSE     TRUE     TRUE     FALSE     FALSE

summary(model_h_locus)$coeff[-1,4] < 0.05

##      h_locus      sex      age      educ      reg eco_insec
##      TRUE      TRUE     FALSE     TRUE     FALSE     FALSE

summary(model_he_eco)$coeff[-1,4] < 0.05

##      he_eco      sex      age      educ      reg eco_insec
##      TRUE      TRUE     FALSE     TRUE     FALSE     FALSE

summary(model_PTV_L)$coeff[-1,4] < 0.05

##      PTV_L      sex      age      educ      reg eco_insec
##      TRUE      TRUE     FALSE     TRUE     FALSE     TRUE

summary(model_PTV_5SM)$coeff[-1,4] < 0.05 #not significant

##      PTV_5SM      sex      age      educ      reg eco_insec
##      FALSE      TRUE     FALSE     TRUE     FALSE     TRUE

summary(model_PTV_BOI)$coeff[-1,4] < 0.05

##      PTV_BOI      sex      age      educ      reg eco_insec
##      TRUE      TRUE     FALSE     TRUE     FALSE     TRUE

summary(model_tr_sci)$coeff[-1,4] < 0.05

##      tr_sci      sex      age      educ      reg eco_insec
##      TRUE      FALSE     TRUE     TRUE     FALSE     FALSE

summary(model_pray)$coeff[-1,4] < 0.05 #not significant

##      pray      sex      age      educ      reg eco_insec
##      FALSE      TRUE     FALSE     TRUE     FALSE     TRUE

summary(model_media)$coeff[-1,4] < 0.05 #not significant

##      media      sex      age      educ      reg eco_insec
##      FALSE      TRUE     FALSE     TRUE     FALSE     TRUE

summary(model_low_comp)$coeff[-1,4] < 0.05

##      low_comp      sex      age      educ      reg eco_insec
##      TRUE      TRUE     FALSE     TRUE     FALSE     TRUE

summary(model_distrust_gov)$coeff[-1,4] < 0.05

## distrust_gov      sex      age      educ      reg      eco
## _insec
##      TRUE      TRUE     FALSE     TRUE     FALSE
FALSE

summary(model_distrust_inst)$coeff[-1,4] < 0.05

```

```

## distrust_inst          sex          age          educ          reg
##           TRUE          TRUE          FALSE          TRUE          FALSE
##      eco_insec
##           FALSE

#vector for significance
Significance = rep(c("Significant"),17)
Significance[3] = "Not significant"
Significance[10] = "Not significant"
Significance[13] = "Not significant"
Significance[14] = "Not significant"

#Gathering strength scores
strength = centMGM %>%
  filter(measure=="Strength") %>%
  dplyr::select(value) %>%
  filter(!row_number() %in% c(15:20))

cent_all = centMGM %>%
  filter(measure=="Strength") %>%
  dplyr::select(value, node) %>%
  mutate_if(is.numeric, ~round(., 2))

#Gathering degree
degree_cent_graph = degree_cent[-c(15:20)]

#Gathering names
names = shortnames
names = names[-c(15:20)]

#Merging into a df1 (Reported in Supplement S1)
scatterplot = data.frame(names, coeff_exp, strength, Significance) %>%
  rename(strength = value)
scatterplot$names = as.factor(scatterplot$names)
scatterplot$coeff_exp = as.numeric(scatterplot$coeff_exp)
scatterplot$strength = as.numeric(scatterplot$strength)
scatterplot$Significance = as.factor(scatterplot$Significance)

#Merging into a df2
scatterplot_degree = data.frame(names, coeff_exp, degree_cent_graph, Significance)
scatterplot_degree$degree_cent_graph = as.numeric(scatterplot_degree$degree_cent_graph)

print(scatterplot_degree)

##           names coeff_exp degree_cent_graph Significance
## 1      vac_bad 3.3883357             10      Significant
## 2      vac_free 5.6708848              4      Significant
## 3    low_worry 1.1215813              6 Not significant
## 4    low_risk  1.3108932              2      Significant
## 5   conspiracy 5.1779681             15      Significant
## 6          nat 1.3207578             15      Significant
## 7    int_locus 0.7773671              8      Significant
## 8 low_col_resp 1.2379600             11      Significant

```

## 9	PTV_L	1.1313915	5	Significant
## 10	PTV_5SM	0.9810478	7	Not significant
## 11	PTV_BOI	1.1476554	6	Significant
## 12	distrust_sci	3.3740932	10	Significant
## 13	pray	0.7902768	15	Not significant
## 14	media	0.9895658	5	Not significant
## 15	low_comp	1.5665293	12	Significant
## 16	distrust_gov	1.4257922	16	Significant
## 17	distrust_inst	1.3790392	9	Significant

Scatterplots

#Strength (Article)

```
graph_strength = ggplot(scatterplot, aes(x=strength, y=coeff_exp, label =
names)) +
  geom_point((aes(color=Significance))) + theme_nice() +
  geom_smooth(method=lm, se = FALSE, linetype = "dotted", color = "#DCDCDC") +
  geom_text_repel(size = 3) +
  xlab("Strength centrality") + ylab("Regression coefficient") +
  scale_fill_discrete(labels=c('label1', 'label2'))
```

```
ggsave(here("Output", "Article", "graph_strength.jpg"), graph_strength, height = 5, width = 8)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation: label
```

```
## i This can happen when ggplot fails to infer the correct grouping structure in
```

```
## the data.
```

```
## i Did you forget to specify a `group` aesthetic or to convert a numerical
```

```
## variable into a factor?
```

#Degree (Supplement S1)

```
graph_degree = ggplot(scatterplot_degree, aes(x=degree_cent_graph, y=coeff_exp, label = names)) +
  geom_point((aes(color=Significance))) + theme_nice() +
  geom_smooth(method=lm, se = FALSE, linetype = "dotted", color = "#DCDCDC") +
  geom_text_repel(size = 3) +
  xlab("Degree centrality") + ylab("Regression coefficient") +
  scale_fill_discrete(labels=c('label1', 'label2'))
```

```
ggsave(here("Output", "Supplement", "graph_degree.jpg"), graph_degree, height = 5, width = 8)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation: label
```

```
## i This can happen when ggplot fails to infer the correct grouping structure in
```

```
## the data.
```

```
## i Did you forget to specify a `group` aesthetic or to convert a numeric
al
## variable into a factor?
```

Final table empirical predictors (Article)

#Df with variables that (1) are tied to hesitancy, (2) are in the backbone

```
final_tab = scatterplot %>%
  mutate(degree = scatterplot_degree$degree_cent_graph) %>%
  add_row(names = "educ", coeff_exp = 1.764247, strength = 1.0869328,
          Significance = "Significant", degree = 10) %>%
  filter(names %in% c("vac_free", "vac_bad", "low_comp", "conspiracy", "nat", "
educ")) %>%
  mutate(edge_weight=c(0.34, 0.95, 0.12, 0.07, 0.34, 0.07)) %>% #adding ed
ge weights
  relocate(Significance, .after = edge_weight)
```

#conversion of type

```
final_tab$names = as.factor(final_tab$names)
final_tab$coeff_exp = as.numeric(final_tab$coeff_exp)
final_tab$strength = as.numeric(final_tab$strength)
final_tab$degree = as.numeric(final_tab$degree)
final_tab$edge_weight = as.numeric(final_tab$edge_weight)
final_tab$Significance = as.factor(final_tab$Significance)
```

#Z scores

```
final_tab_z = final_tab %>%
  mutate(coeff_exp = (coeff_exp - mean(coeff_exp))/sd(coeff_exp),
          strength = (strength - mean(strength))/sd(strength),
          degree = (degree - mean(degree))/sd(degree),
          edge_weight = (edge_weight - mean(edge_weight))/sd(edge_weight))
%>%
  mutate_if(is.numeric, ~round(., 2)) %>%
  arrange(desc(coeff_exp)) %>%
  dplyr::select(names, coeff_exp, edge_weight, strength, degree, Significa
nce)
```

final_tab

##	names	coeff_exp	strength	degree	edge_weight	Significance
## 1	vac_bad	3.388336	1.612156	10	0.34	Significant
## 2	vac_free	5.670885	1.340933	4	0.95	Significant
## 3	conspiracy	5.177968	1.833237	15	0.12	Significant
## 4	nat	1.320758	1.272407	15	0.07	Significant
## 5	low_comp	1.566529	1.470828	12	0.34	Significant
## 6	educ	1.764247	1.086933	10	0.07	Significant

final_tab_z

##	names	coeff_exp	edge_weight	strength	degree	Significance
## 1	vac_free	1.32	1.89	-0.36	-1.71	Significant
## 2	conspiracy	1.06	-0.58	1.51	0.98	Significant
## 3	vac_bad	0.13	0.07	0.67	-0.24	Significant
## 4	educ	-0.72	-0.73	-1.32	-0.24	Significant

```
## 5 low_comp -0.83 0.07 0.13 0.24 Significant
## 6 nat -0.96 -0.73 -0.62 0.98 Significant
```

#Table strength and degree centrality of each node (Supplement)

```
table_cent = cent_all
table_cent[,3]= degree_data[,2]
table_cent = table_cent %>% dplyr::select(node, value, V3) %>%
  rename(Strength = value, Degree = V3)
```

table_cent

```
##
## node Strength Degree
## 1 Vaccine bad for health 1.61 10
## 2 Against mandatory vaccination 1.34 4
## 3 Low worry about infection 0.75 6
## 4 Low risk perception 0.28 2
## 5 Endorsing conspiracy theories 1.83 15
## 6 Trust in alternative medicine 1.27 15
## 7 Internal health locus of control 0.54 8
## 8 Low sense of collective responsibility 0.79 11
## 9 Propensity to vote for L 1.12 5
## 10 Propensity to vote for 5SM 0.74 7
## 11 Propensity to vote for BOI 1.01 6
## 12 Distrust of science 1.22 10
## 13 Religion, pray 1.49 15
## 14 Digital media diet 0.48 5
## 15 Sex, female 1.01 10
## 16 Age, young 1.24 11
## 17 Educational level, low 1.09 10
## 18 Living in a rural region 0.47 9
## 19 Economic insecurity 0.64 9
## 20 Vaccine hesitancy 2.13 7
## 21 Low compliance with preventive behaviors 1.47 12
## 22 Disapproval of Government 1.50 16
## 23 Distrust of institutions 1.20 9
```

Output

Export heavy objects

```
#communities
saveRDS(CommunityStabTotal, here("Input", "CommunityStabTotal.rds"))
#bootnet
saveRDS(edgeacc, here("Input", "edgeacc.rds"))
saveRDS(centstab, here("Input", "centstab.rds"))
#final tab
saveRDS(final_tab, here("Input", "final_tab.rds"))
saveRDS(final_tab_z, here("Input", "final_tab_z.rds"))
```

Edge weight to excel (Supplement S1)

```
##only upper triangle without edge weights 0 for readability
EdgeWeight_Total_half<-upper.triangle(inputGraphMGM)
EdgeWeight_Total_half[EdgeWeight_Total_half == 0] <- NA
EdgeWeightsExcel_half<- list("mgm" = EdgeWeight_Total_half)
write.xlsx(EdgeWeightsExcel_half, "../Output/Supplement/EdgeWeightsExcel_h
```

```
alf.xlsx",  
  colWidths = "auto", rowNames = TRUE)
```